

Assignment

1

Student Name
Ahmed Walid (4231011)

Teacher Name
Eng./ Ethar Ibrahim

Q1. Explain the difference between Machine Learning and Deep Learning

- Machine Learning (ML) is a branch of artificial intelligence that allows systems to learn from data and improve their performance without being explicitly programmed. It often requires feature engineering and works well with smaller datasets.
- Deep Learning (DL) is a subset of Machine Learning that uses neural networks with multiple hidden layers to learn complex patterns automatically. It works best with large datasets and requires less manual feature engineering.

Q2. Why is Deep Learning considered a subset of Machine Learning?

Deep Learning is considered a subset of Machine Learning because it follows the same principle of learning from data. However, it uses advanced techniques such as deep neural networks to automatically extract features and model complex relationships, which makes it a more specialized approach within Machine Learning.

Q3. List any two reasons why PyTorch is widely used in deep learning applications

1. PyTorch provides dynamic computation graphs, which makes it easier to debug and experiment.
2. It supports GPU acceleration, which speeds up training of deep learning models.

Q4. What is a neuron in a neural network?

Briefly explain how it works.

A neuron is the basic unit of a neural network. It takes input, multiplies them by weights, adds a bias, and then passes the result through an activation function to produce an output.

$$z = \sum w_i x_i + b$$

$$\text{output} = f(z)$$

Q5. What is the role of an activation function in a neural network?

The activation function introduces non-linearity into the neural network. It helps the model learn complex patterns and relationships in data. Without activation functions, the network would behave like a simple linear model and would not be able to solve complex problems.
